

When Does Raw AGOP Explain Learned Features?

A Weighted Stationarity View of Two-Layer Feature Learning

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Abstract

The Neural Feature Ansatz (NFA) observes that a trained network’s first-layer feature matrix tracks an average gradient outer product (AGOP). We ask what the optimization itself implies in a two-layer model with square loss and L_2 regularization on the first layer. Expanding the first-layer stationarity equation gives an exact, finite-sample answer: the stable convergence-side object is not raw AGOP but a *residual-weighted* AGOP, $H^2 \approx \kappa_{\text{eff}} \tilde{G}$, and $\tilde{G}$ is exactly the covariance of the input gradients of the loss. Raw AGOP is then a derived, regime-dependent shadow of this law, gated by two checkable links: a scalar *beta bridge* and *high-gain pair closure*. The weighted-law error is an exact sum of a pushed pair-isotropy term and a stationarity defect, and we show the pushed term is what matters: a large global pair defect can coexist with a tiny pushed error. We classify how each link fails, prove the two links are logically independent, and report a synthetic-network study: across nine teacher-student families and five seeds, the weighted law is empirically robust, even to constructions built to break high-gain pair closure, and degrades only under persistent-residual outlier structure, which also bends the beta link.

1. Introduction

The NFA, posited by Radhakrishnan et al. [?] and central to recursive feature machines, states that the learned first-layer geometry $H := B^\top B$ of a trained network is proportional to a power of the AGOP $G := \frac{1}{n} \sum_i \nabla f(x_i) \nabla f(x_i)^\top$. Empirically the relation is striking, but a general nonlinear theory is open: for deep networks, NFA-type statements have been proved only in linear regimes, with nonlinear counterexamples [?]. The Features-at-Convergence viewpoint (FACT) [?] reframes the question, deriving a self-consistent feature relation from first-order optimality rather than fitting AGOP after training, and explaining when NFA holds or fails.

This report takes that viewpoint literally in the simplest nonlinear model that has the structure: a two-layer network trained with square loss and ridge on the first layer. Expanding the first-layer stationarity equation gives a clean answer with three pieces. **(1) The natural object is residual-weighted, not raw:** stationarity yields a deterministic identity $H^2 - \kappa_{\text{eff}} \tilde{G} = (\lambda^2 n^2)^{-1} B^\top (S - d_{\text{eff}} T) B + B^\top E_{\text{stat}} B$, where $\tilde{G}$ is the residual-weighted AGOP, which equals the covariance of the input gradients of the loss, so the convergence-side object is derived rather than assumed. **(2) Raw AGOP needs two extra links:** a scalar *beta bridge* ($\tilde{M} \approx \beta M$) converts $\tilde{G}$ into G , and the weighted law itself requires *high-gain pair closure*. We give exact identities for both and the failure taxonomy they imply, and a 2-dimensional argument shows the two links are logically independent, so no theorem with hypotheses limited to stationarity and finite-dimensional algebra can imply a universal raw NFA law. **(3) The right pair diagnostic is not the global one:** the pushed pair error $B^\top (S - d_{\text{eff}} T) B$ factors through a stationarity-induced gain map, so a large global pair-isotropy defect can be conservative. The contribution is this finite-sample decomposition, the diagnostics that read those links off any checkpoint, and a synthetic study mapping which link fails where. Closely related concurrent work studies $B^\top B$ as a feature-dynamics object via a “Virtual Covariance” [?], and feature-learning *dynamics* in two-layer networks from small initialization is studied separately [?].

2. Setup and first-order identities

The model is $f(x) = a^\top \phi(Bx)$ with $B \in \mathbb{R}^{m \times d}$, $a \in \mathbb{R}^m$, smooth activation ϕ . With training data $(x_i, y_i)_{i=1}^n$, residuals $r_i := f(x_i) - y_i$, and hidden-derivative features $q_i := a \odot \phi'(Bx_i)$, write $X = [x_1, \dots, x_n]$, $Q = [q_1, \dots, q_n]$, $R = \text{diag}(r_1, \dots, r_n)$. By the chain rule $\nabla f(x_i) = B^\top q_i$. Define

$$M := \frac{1}{n} Q Q^\top, \quad \tilde{M} := \frac{1}{n} Q R^2 Q^\top, \quad G := B^\top M B, \quad \tilde{G} := B^\top \tilde{M} B = \frac{1}{n} \sum_i r_i^2 \nabla f(x_i) \nabla f(x_i)^\top.$$

Since the per-sample loss is $\ell_i = \frac{1}{2} r_i^2$ and $\nabla_x \ell_i = r_i \nabla f(x_i)$, the weighted object is exactly the covariance of the input gradients of the loss: $\tilde{G} = \frac{1}{n} \sum_i \nabla_x \ell_i \nabla_x \ell_i^\top$. Training minimizes $\mathcal{L}(B, a) = \frac{1}{2n} \sum_i r_i^2 + \frac{\lambda}{2} \|B\|_F^2$, so $\nabla_B \mathcal{L} = \frac{1}{n} Q R X^\top + \lambda B$, and at exact B -stationarity

$$B = -\frac{1}{\lambda n} Q R X^\top, \quad \text{with defect} \quad Z := \lambda n B + Q R X^\top.$$

The residual weighting is not a modeling choice. It is the derivative of the square loss, and it enters B automatically.

3. The deterministic weighted square law

Let $T := QR^2Q^\top = n\widetilde{M}$, $S := QRX^\top XRQ^\top$, $d_{\text{eff}} := \langle S, T \rangle_F / \|T\|_F^2$ (the best Frobenius scalar fitting $S \approx dT$), and $\kappa_{\text{eff}} := d_{\text{eff}} / (\lambda^2 n)$. From $\lambda n B = -QRX^\top + Z$ we get $BB^\top = (\lambda^2 n^2)^{-1} S + E_{\text{stat}}$ with $E_{\text{stat}} := (\lambda^2 n^2)^{-1} (-QRX^\top Z^\top - ZX RQ^\top + ZZ^\top)$. Multiplying by $B^\top(\cdot)B$ and using $H^2 = B^\top(BB^\top)B$ gives the central result.

Proposition 1 (Deterministic weighted law). *With the notation above,*

$$H^2 - \kappa_{\text{eff}} \widetilde{G} = \frac{1}{\lambda^2 n^2} B^\top (S - d_{\text{eff}} T) B + B^\top E_{\text{stat}} B, \quad \text{hence} \quad \|H^2 - \kappa_{\text{eff}} \widetilde{G}\|_{\text{op}} \leq \|B\|_{\text{op}}^2 \left(\frac{\|T\|_{\text{op}}}{\lambda^2 n^2} \mathcal{A}_{\text{pair}} + \|E_{\text{stat}}\|_{\text{op}} \right),$$

where $\mathcal{A}_{\text{pair}} := \|T^{\dagger/2} (S - d_{\text{eff}} T) T^{\dagger/2}\|_{\text{op}}$. In particular, if $Z = 0$ and $S = d_{\text{eff}} T$ on $\text{range}(T)$, then $H^2 = \kappa_{\text{eff}} \widetilde{G}$ exactly.

The pushed pair error is the operative quantity. The bound uses $\mathcal{A}_{\text{pair}}$, the worst pair-isotropy error over the whole support of T , but the law only sees the term $B^\top (S - d_{\text{eff}} T) B$. Let $QR = U\Sigma V^\top$ be the thin SVD ($U \in \mathbb{R}^{m \times r}$ in hidden space, $V \in \mathbb{R}^{n \times r}$ in sample space), and set $H_X := V^\top X^\top X V$, $F_X := H_X - d_{\text{eff}} I_r$, $G_{\text{stat}} := \Sigma^2 V^\top X^\top$. Then $S - d_{\text{eff}} T = U \Sigma F_X \Sigma U^\top$, $\mathcal{A}_{\text{pair}} = \|F_X\|_{\text{op}}$, and at exact stationarity, since $B = -(\lambda n)^{-1} U \Sigma V^\top X^\top$,

$$B^\top (S - d_{\text{eff}} T) B = \frac{1}{\lambda^2 n^2} G_{\text{stat}}^\top F_X G_{\text{stat}}.$$

So F_X is the pair-isotropy defect of the input geometry $X^\top X$ restricted to the sample combinations the residual-weighted hidden gradients use, G_{stat} weights each such direction by σ_k^2 and projects it through $X^\top$, and the weighted law breaks only when F_X is large on the high-gain directions of G_{stat} . A large global $\mathcal{A}_{\text{pair}} = \|F_X\|_{\text{op}}$ with all the defect on low-gain directions costs almost nothing. A sufficient probabilistic mechanism: if a high-gain right-singular subspace $W = VP_{\text{hi}}$ is fixed or independent of Gaussian X , then $W^\top X^\top X W \approx dI$ by standard Wishart concentration, so F_X is small there. The catch is that the learned W depends on X , so this is a sufficient condition, not a proof for trained networks (Sec. ??). Upgrading it requires an adaptive-concentration argument over a low-complexity class of learnable subspaces.

4. From weighted to raw AGOP, and the failure taxonomy

Raw AGOP needs a scalar bridge $\widetilde{M} \approx \beta M$. The Frobenius best-fit one is $\beta_{\text{fit}} := \langle \widetilde{M}, M \rangle_F / \|M\|_F^2$. Using $\langle q_i q_i^\top, q_j q_j^\top \rangle_F = (q_i^\top q_j)^2$,

$$\beta_{\text{fit}} = \frac{\sum_{i,j} r_i^2 (q_i^\top q_j)^2}{\sum_{i,j} (q_i^\top q_j)^2} = \frac{1}{n} \sum_i \ell_i r_i^2, \quad \ell_i := \frac{\sum_j (q_i^\top q_j)^2}{\frac{1}{n} \sum_k \sum_j (q_k^\top q_j)^2}, \quad \frac{1}{n} \sum_i \ell_i = 1,$$

i.e. β_{fit} is a hidden-gradient-leverage-weighted average of the squared residuals, with $\min_i r_i^2 \leq \beta_{\text{fit}} \leq \max_i r_i^2$. Hence

$$\frac{\beta_{\text{fit}}}{\bar{r}^2} - 1 = \text{Corr}_n(\ell, r^2) \text{CV}_n(\ell) \text{CV}_n(r^2), \quad \bar{r}^2 := \frac{1}{n} \sum_i r_i^2,$$

so the beta bridge tracks mean residual energy precisely when hidden-gradient leverage and residual energy are weakly correlated. Combining with Prop. ???: if $\|H^2 - \kappa_{\text{eff}} \widetilde{G}\|_{\text{op}} \leq E_w$ and $\|\widetilde{G} - \beta_{\text{fit}} G\|_{\text{op}} \leq E_\beta$, then $\|H^2 - \kappa_{\text{eff}} \beta_{\text{fit}} G\|_{\text{op}} \leq E_w + \kappa_{\text{eff}} E_\beta$, and solving for G divides by $|\beta_{\text{fit}}|$. So the raw conversion is well conditioned only away from interpolation, where β_{fit} is bounded away from zero. This is the *two-regime picture*: $H^2 \approx \kappa_{\text{eff}} \widetilde{G}$ is the stable late object, while $H^2 \approx \kappa_{\text{eff}} \beta_{\text{fit}} G$ (raw-AGOP-like) is visible only in an intermediate regime.

Failure taxonomy. Three distinct mechanisms break NFA-type behavior. (i) *Beta failure*: $\text{Corr}_n(\ell, r^2)$ is large, so β_{fit} deviates from $\bar{r}^2$, the raw law is biased, but the weighted law is unaffected. (ii) *High-gain pair failure*: F_X is large on the high-gain directions of G_{stat} , so $G_{\text{stat}}^\top F_X G_{\text{stat}}$ is large and the weighted law itself fails. (iii) *Late-regime collapse*: near interpolation $\beta_{\text{fit}} \rightarrow 0$, so the raw conversion is ill-conditioned even when the weighted law is exact. A fourth case is benign: $\mathcal{A}_{\text{pair}}$ large but the pushed error small (the conservative-diagnostic case). The two links are logically independent: there are explicit 2-dimensional stationary instances ($n = 2$, $\lambda n = 1$) where (a) the weighted law is exact while $\widetilde{M} = \beta M$ has no scalar solution, (b) the beta bridge is exact while a high-gain pair defect breaks the weighted law, and (c) $\mathcal{A}_{\text{pair}}$ is order one while the pushed error is $O(\varepsilon^2)$. Consequently:

Corollary 1. *No theorem whose only hypotheses are exact B-stationarity and finite-dimensional algebra can imply a universal exact raw scalar NFA law $H \approx cG^\alpha$. The two bridge conditions (beta stability and high-gain pair closure) are required.*

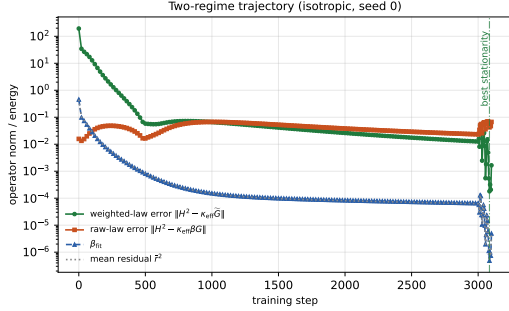


Figure 1: Two-regime trajectory (isotropic, one seed). Green: weighted-law error. Red: raw-AGOP error. Blue: β_{fit} . The weighted-law error decreases into stationarity, the raw error bottoms out in an intermediate window, and β_{fit} collapses with the mean residual energy near interpolation ($\min_i r_i^2 \leq \beta_{\text{fit}} \leq \max_i r_i^2$). The step at the gradient-descent-to-L-BFGS hand-off is a real regime change, not a plotting artifact.

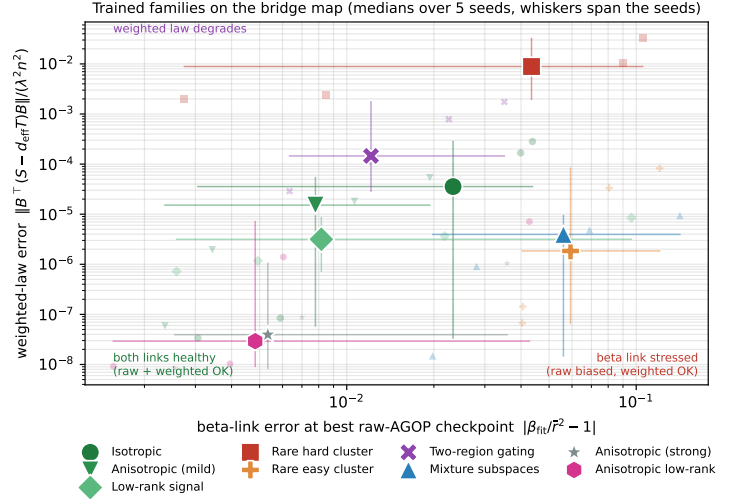


Figure 2: Trained families on the bridge map (5 seeds, markers at family medians, whiskers span the seeds). x : beta-link error $|\beta_{\text{fit}}/\bar{r}^2 - 1|$ at the best raw-AGOP checkpoint. y : weighted-law residual $\|B^T(S - d_{\text{eff}}T)B\|/(\lambda^2 n^2)$ at the best-stationarity checkpoint. The positive families and the two pair-geometry attacks (strong anisotropy, anisotropic low-rank) sit in the bottom-left healthy cluster. The mixture and rare-easy families move right (beta link stressed, weighted law fine), and only the rare hard cluster moves up by orders of magnitude (two-region gating, mildly).

5. Experiments

Protocol. All data are synthetic teacher-student regression. For each run, a fixed random two-layer teacher $f_*(x) = a_*^T \phi(B_*x)$ labels $n = 128$ inputs $x_i \in \mathbb{R}^{256}$ drawn from a chosen distribution (clean labels), and a fresh two-layer student ($m = 16$ hidden units) is trained on those pairs with square loss plus ridge on B ($\lambda = 10^{-2}$), using gradient descent followed by L-BFGS refinement, in double precision, logging the diagnostics at roughly 170 checkpoints. We run five random seeds per family. At each checkpoint we evaluate exactly the objects of Prop. ?? and Sec. 4: the weighted residual $\|H^2 - \kappa_{\text{eff}}\tilde{G}\|_{\text{op}}$, the theorem bound, $\beta_{\text{fit}}/\bar{r}^2$ and $\text{Corr}_n(\ell, r^2)$, the global $\mathcal{A}_{\text{pair}}$ versus the pushed pair error $\|B^T(S - d_{\text{eff}}T)B\|_{\text{op}}/(\lambda^2 n^2)$, and the stationarity defect. Quantities are read at the best-stationarity checkpoint (smallest $\|BB^T - B_{\text{stat}}B_{\text{stat}}^T\|_{\text{op}}$) unless noted. The data families are three positive controls (isotropic Gaussian, anisotropic Gaussian, low-rank signal plus isotropic noise) and six adversarial families built to attack a specific link (rare hard and rare easy high-leverage clusters, two-region gating, mixture of subspaces, strongly anisotropic Gaussian, and a low-rank signal with a steeply anisotropic within-subspace covariance).

Positive families. At best-stationarity checkpoints the weighted law is accurate and within its deterministic bound: mean weighted residual $\approx 3.6 \times 10^{-3}$, 1.9×10^{-3} , 6.4×10^{-4} for the isotropic, anisotropic, and low-rank-signal families, with mean theorem-bound ratios ≈ 0.15 , 0.090 , 0.099 . The beta bridge is benign: $\beta_{\text{fit}}/\bar{r}^2$ averages 0.99 , 1.00 , 1.02 across seeds, with small spread. And the global pair diagnostic is conservative: $\mathcal{A}_{\text{pair}}$ averages roughly 80 to 280 while the pushed pair error averages $\approx 5.6 \times 10^{-5}$, 1.1×10^{-5} , 2.5×10^{-6} , five to seven orders of magnitude smaller, exactly the case the SVD factorization predicts. Figure ?? shows the trajectory: a single end-of-training correlation number hides the structure that the decomposition makes explicit.

Trained bridge map and a robustness finding. Figure ?? places all nine families on axes given by the two links. The headline observation is that the weighted law is hard to break. Two families built deliberately to attack high-gain pair closure, a strongly anisotropic Gaussian (covariance spectrum spanning four orders of magnitude) and a low-rank signal with a steeply anisotropic within-subspace covariance, both land in the healthy bottom-left cluster with the positive controls, with pushed pair errors near 10^{-7} even though their global $\mathcal{A}_{\text{pair}}$ and even their high-gain closure norms are sizeable. The reason is structural: near stationarity $B = -(\lambda n)^{-1}QRX^T$, so the only directions that enter the pushed error are the ones the residual-weighted hidden gradients span, and the network self-aligns those directions with the input geometry, keeping $S \approx d_{\text{eff}}T$ on exactly the directions B sees. The

mixture-of-subspaces and rare-easy families instead move right: they stress the beta link ($|\beta_{\text{fit}}/\bar{r}^2 - 1|$ rises to about 0.05, raw AGOP biased), while the weighted law stays fine. The only family that moves up by orders of magnitude is the rare hard cluster (and, mildly, two-region gating): there, amplified outlier labels keep the residuals of a small set large persistently, sustaining a misaligned QR that the network cannot iron out at stationarity, and that same persistence also bends the beta link, so it is a combined failure rather than a pure pair one. In these runs the weighted law degrades only under persistent-residual outlier structure.

6. Discussion

Relation to recent work. AGOP and recursive feature machines [?] make AGOP a central object of feature learning, and FACT [?] derives convergence-side feature relations from first-order optimality and explains when NFA holds. The present work is the concrete two-layer square-loss specialization of that mechanism, with the novel part being the bridge decomposition: identifying $\tilde{G}$ as the loss-gradient covariance, isolating the beta and high-gain-pair links, showing the global pair diagnostic is conservative, and mapping which link fails where. Deep-NFA theory proves NFA in linear regimes with nonlinear counterexamples [?], consistent with Corollary ???. Concurrent work studies $B^\top B$ as a feature-dynamics object via a Virtual Covariance and a Feature Learning Equation [?], and relating that quantity to $\tilde{G}$ is a natural next step. Feature-learning dynamics from small initialization is treated by alternating gradient flows [?], a candidate tool for the open dynamics direction below.

Limitations. The model is two-layer with square loss and ridge on B only, and the families are synthetic. The deterministic identity assumes exact stationarity. We report near it and $\log E_{\text{stat}}$ so its contribution is visible, and the theorem bound uses the conservative $\mathcal{A}_{\text{pair}}$, so the bound ratio is well below one rather than tight. The high-gain pair-closure result is conditional (a frozen-subspace concentration mechanism), not a proof for trained networks. The trained map is a guiding map across nine families, not a deliberate grid sweep over the leverage-residual correlation crossed with high-gain anisotropy, and per-seed spread is wide while family medians separate cleanly.

Open directions. (i) Derive the beta decorrelation (or high-gain pair closure) from training dynamics rather than assuming it. The relevant empirical probe, whether high hidden-gradient leverage predicts faster residual decay, is logged. (ii) Isolate a pure high-gain pair failure: every pair-geometry attack so far self-aligns near stationarity. (iii) A full deliberate phase diagram with samplewise diagnostics, and one non-synthetic dataset. (iv) Extend the law beyond two layers by replacing B and q_i with layerwise Jacobian factors.

7. Conclusion

In a two-layer model trained with square loss and ridge on the first layer, first-order stationarity does not predict raw AGOP. It predicts a residual-weighted AGOP law $H^2 \approx \kappa_{\text{eff}} \tilde{G}$, with $\tilde{G}$ equal to the covariance of the input gradients of the loss, and with an exact error split into a pushed pair-isotropy term and a stationarity defect. Raw AGOP is a derived, regime-dependent consequence, gated by a beta bridge and high-gain pair closure, which are logically independent and individually checkable, so no universal raw NFA law follows from stationarity alone. Empirically, the weighted law is robust: across synthetic families it degrades only under persistent-residual outlier structure, and even that case is a combined beta-and-pair failure rather than a pure pair one. The takeaway is that AGOP/NFA-type relations are governed by beta conditioning and high-gain pair closure, and the residual-weighted law is the stable convergence-side object.

References

- [1] A. Radhakrishnan, D. Beaglehole, P. Pandit, M. Belkin. Mechanism for feature learning in neural networks and backpropagation-free machine learning models. *Science*, 2024.
- [2] E. Boix-Adsera, N. Mallinar, J. B. Simon, M. Belkin. The Features at Convergence Theorem: a first-principles alternative to the Neural Feature Ansatz for how networks learn representations. arXiv:2507.05644, 2025.
- [3] E. Tansley, E. Massart, C. Cartis. On the Neural Feature Ansatz for Deep Neural Networks. arXiv:2510.15563, 2025.
- [4] T. Cha, D. Beaglehole, A. Radhakrishnan, D. Lee. The Weight Gram Matrix Captures Sequential Feature Linearization in Deep Networks. arXiv:2605.06258, 2026.
- [5] D. Kunin, G. L. Marchetti, F. Chen, D. Karkada, J. B. Simon, M. R. DeWeese, S. Ganguli, N. Miolane. Alternating Gradient Flows: A Theory of Feature Learning in Two-layer Neural Networks. arXiv:2506.06489, 2025.